Solutions to Midterm

Problem 1

Solution.

(1) $\|x\|_2 = \sqrt{\sum_{i=1}^{n} x_i^2}$

$\|x\|_1 = \sum_{i=1}^{n} |x_i|$

$\|x\|_\infty = \max_{1\le i\le n} |x_i|$

(2) $\|x\|_2^2 = \sum_{i=1}^{n} x_i^2 \le \max_{1\le i\le n} |x_i| \cdot (\sum_{i=1}^{n} |x_i|) = \|x\|_1\|x\|_\infty$

Problem 2

Solution.
Let $x = [x_1, x_2, \cdots, x_n]^T$ and $y = [y_1, y_2, \cdots, y_n]^T$. we have

$$\|xy^T\|_F = \sqrt{\sum_{i=1}^{n}\sum_{j=1}^{n}(x_i y_i)^2} = \sqrt{\sum_{i=1}^{n} x_i^2 \sum_{j=1}^{n} y_i^2} = \|x\|_2\|y\|_2$$

and

$$\|xy^T\|_2 = \left\| \frac{x}{\|x\|_2}\|x\|_2 \cdot \|y\|_2 \frac{y^T}{\|y\|_2} \right\|_2 = \|x\|_2\|y\|_2 \cdot \left\| \frac{x}{\|x\|_2}\frac{y^T}{\|y\|_2} \right\|_2 = \|x\|_2\|y\|_2.$$

Thus

$$\|xy^T\|_F = \|xy^T\|_2 = \|x\|_2\|y\|_2.$$

Problem 3

Solution.

$$
\begin{aligned}
A = \begin{bmatrix} 16 & 4 & 4 & -4 \\ 4 & 10 & 4 & 2 \\ 4 & 4 & 6 & -2 \\ -4 & 2 & -2 & 4 \end{bmatrix} &= \begin{bmatrix} 4 & & & \\ 1 & 1 & & \\ 1 & & 1 & \\ -1 & & & 1 \end{bmatrix} \begin{bmatrix} 1 & & & \\ & 9 & 3 & 3 \\ & 3 & 5 & -1 \\ & 3 & -1 & 3 \end{bmatrix} \begin{bmatrix} 4 & 1 & 1 & -1 \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix} \\
&= \begin{bmatrix} 4 & & & \\ 1 & 3 & & \\ 1 & 1 & 1 & \\ -1 & 1 & & 1 \end{bmatrix} \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 4 & -2 \\ & & -2 & -2 \end{bmatrix} \begin{bmatrix} 4 & 1 & 1 & -1 \\ & 3 & 1 & 1 \\ & & 1 & \\ & & & 1 \end{bmatrix} \\
&= \begin{bmatrix} 4 & & & \\ 1 & 3 & & \\ 1 & 1 & 2 & \\ -1 & 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix} \begin{bmatrix} 4 & 1 & 1 & -1 \\ & 3 & 1 & 1 \\ & & 2 & -1 \\ & & & 1 \end{bmatrix} \\
&= \begin{bmatrix} 4 & & & \\ 1 & 3 & & \\ 1 & 1 & 2 & \\ -1 & 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 4 & 1 & 1 & -1 \\ & 3 & 1 & 1 \\ & & 2 & -1 \\ & & & 1 \end{bmatrix}.
\end{aligned}
$$

Thus

$$
L = \begin{bmatrix} 4 & & & \\ 1 & 3 & & \\ 1 & 1 & 2 & \\ -1 & 1 & -1 & 1 \end{bmatrix}, L^T = \begin{bmatrix} 4 & 1 & 1 & -1 \\ & 3 & 1 & 1 \\ & & 2 & -1 \\ & & & 1 \end{bmatrix}.
$$

Note that $Ax = b$ is equivalent to $LL^Tx = b$. Let $L^Tx = y$, then $Ly = b$, we can calculate that $y = [8, 6, 3, -1]^T$. From $L^Tx = y$, we can obtain that $x = [1, 2, 1, -1]^T$.

Problem 4

Solution.

$$A = \begin{bmatrix} -1 & -1 & 1 \\ 1 & 3 & 3 \\ -1 & -1 & 5 \\ 1 & 3 & 7 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} & -1 & 1 \\ \frac{1}{2} & 3 & 3 \\ -\frac{1}{2} & -1 & 5 \\ \frac{1}{2} & 3 & 7 \end{bmatrix} \begin{bmatrix} 2 & & \\ & 1 & \\ & & 1 \end{bmatrix}$$
$$= \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} & 1 \\ \frac{1}{2} & \frac{1}{2} & 3 \\ -\frac{1}{2} & \frac{1}{2} & 5 \\ \frac{1}{2} & \frac{1}{2} & 7 \end{bmatrix} \begin{bmatrix} 2 & 4 & \\ & 2 & \\ & & 1 \end{bmatrix}$$
$$= \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 2 & 4 & 2 \\ & 2 & 8 \\ & & 4 \end{bmatrix}.$$

Thus

$$Q = \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix}, R = \begin{bmatrix} 2 & 4 & 2 \\ & 2 & 8 \\ & & 4 \end{bmatrix}.$$

Problem 5

Solution.
Let $x_1 = [2, 1, 1]^T, y_1 = \|x_1\|_2 e_1 = [3, 0, 0]^T$ and $v_1 = x_1 - y_1 = [-1, 2, 1]^T$. Then

$$H_1 = I - 2\frac{v_1 v_1^T}{v_1^T v_1} = \begin{bmatrix} \frac{2}{3} & \frac{2}{3} & \frac{1}{3} \\ \frac{2}{3} & -\frac{1}{3} & -\frac{2}{3} \\ \frac{1}{3} & -\frac{2}{3} & \frac{2}{3} \end{bmatrix}.$$

Let $Q_1 = \begin{bmatrix} 1 & \\ & H_1 \end{bmatrix}$. It's clear that Q_1 is symmetric and orthogonal. We can obtain that

$$Q_1 A Q_1 = \begin{bmatrix} -1 & 3 & 0 & 0 \\ 3 & \frac{1}{3} & \frac{5}{3} & 0 \\ 0 & \frac{5}{3} & 3 & \frac{2}{3} \\ 0 & 0 & \frac{2}{3} & -\frac{7}{3} \end{bmatrix},$$

which is a tridiagonal matrix obviously. Let $Q_2 = I$, which can satisfy $Q_2 Q_1 A Q_1 Q_2$ is tridiagonal.

Problem 6

Solution.

(1) One of the eigenvalue of A is 0, corresponding to $n-1$ eigenvectors $z_1, \cdots, z_{n-1}$, which are independent such that $y^T z_i = 0$ for $i = 1, \cdots, n-1$. Another is 1, whose corresponding eigenvector is x.

(2)
- Power Iteration
  Given v_0 with $v_0 = a_1 z_1 + \cdots + a_{n-1} z_{n-1} + a_n x$ and $\|v_0\|_2 = 1$
  ($z_1, \cdots, z_{n-1}, x$ are given in (1).)
  for $k = 1, 2, \cdots$
  $w = (A - \frac{1}{2}I)v_{k-1}$
  $v_k = w/\|w\|_2$
  $\lambda_k = v_k^T (A - \frac{1}{2}I)v_k$
  end
- $Ax = \lambda x \Rightarrow Ax - \frac{1}{2}x = \lambda x - \frac{1}{2}x \Rightarrow (A - \frac{1}{2}I)x = (\lambda - \frac{1}{2})x$.
  Thus the eigenvalues of $A - \frac{1}{2}I$ are $\frac{1}{2}$ and $-\frac{1}{2}$, which have the same absolute value. It follows that $\left|\frac{\lambda_2}{\lambda_1}\right|^k = \left|\frac{\lambda_3}{\lambda_1}\right|^k \cdots = \left|\frac{\lambda_n}{\lambda_1}\right|^k = 1$. Therefore, the algorithm will not converge.

Problem 7

Solution.
Let $|A - \lambda I| = 0$, that is

$$\begin{vmatrix} \lambda - 3 & -1 & \\ -1 & \lambda - 3 & \\ & & \lambda - 1 \end{vmatrix} = 0.$$

Denote the eigenvalues of A by $\lambda_1, \lambda_2, \lambda_3$, then we can solve $\lambda_1 = 1, \lambda_2 = 2, \lambda_3 = 4$. If we want to find the second largest eigenvalue of A, the value of the shift μ should satisfy

$$\begin{cases} |\lambda_2 - \mu| < |\lambda_1 - \mu| \\ |\lambda_2 - \mu| < |\lambda_3 - \mu|. \end{cases}$$

Therefore, $\mu \in (\frac{3}{2}, 2) \cup (2, 3)$.

Problem 8

Solution.

(1) From the weighted graph, we can obtain

$$D = \begin{bmatrix} 0.4 & 0 & 0 & 0 & 0 \\ 0 & 0.2 & 0 & 0 & 0 \\ 0 & 0 & 0.4 & 0 & 0 \\ 0 & 0 & 0 & 0.2 & 0 \\ 0 & 0 & 0 & 0 & 0.2 \end{bmatrix}, W = \begin{bmatrix} 0 & 0.1 & 0.3 & 0 & 0 \\ 0.1 & 0 & 0.1 & 0 & 0 \\ 0.3 & 0.1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.2 \\ 0 & 0 & 0 & 0.2 & 0 \end{bmatrix}.$$

Thus

$$L = D - W = \begin{bmatrix} 0.4 & -0.1 & -0.3 & 0 & 0 \\ -0.1 & 0.2 & -0.1 & 0 & 0 \\ -0.3 & -0.1 & 0.4 & 0 & 0 \\ 0 & 0 & 0 & 0.2 & -0.2 \\ 0 & 0 & 0 & -0.2 & 0.2 \end{bmatrix}.$$

(2) Let $L_1 = \begin{bmatrix} 0.4 & -0.1 & -0.3 \\ -0.1 & 0.2 & -0.1 \\ -0.3 & -0.1 & 0.4 \end{bmatrix}$ and $L_2 = \begin{bmatrix} 0.2 & -0.2 \\ -0.2 & 0.2 \end{bmatrix}$. Easy to see L_1 and L_2 are positive semidefinite. Then we can obtain that the smallest eigenvalue of L_1 and L_2 both equals 0 where the corresponding eigenvectors are $[1, 1, 1]^T$ and $[1, 1]^T$, respectively. Since $L = \begin{bmatrix} L_1 & 0 \\ 0 & L_2 \end{bmatrix}$, L have the same eigenvalues as L_1 and L_2. It follows that the smallest eigenvalue of L is 0. Let $y_1 = [1, 1, 1, 0, 0]^T$ and $y_2 = [0, 0, 0, 1, 1]^T$. We have $Ly_1 = 0y_1$ and $Ly_2 = 0y_2$. Therefore, 0 is the smallest eigenvalue with multiplicity 2 and the associated eigenvectors are $ay_1 + by_2$(with a and b not both 0).